

Study Quick Guide

1. Overview

- This study includes 4 conditions and 3 tasks.
- Conditions: Macro - Near Head, Macro - Side Of Body, Micro - Near Head, Micro - Side Of Body
- Tasks: Placement, Rotation, Scaling
- The condition order may vary depending on the Participant ID.
- In both Macro and Micro, submission is always triple tap for every task.

2. Procedure

1. On the start screen, enter your assigned Participant ID.
2. If needed, use Edit to change it, then press Continue.
3. In the runtime scene, when you see Scan the QR code to begin., scan the HoloLens workspace QR code.
4. At the start of each condition, the condition name appears. Listen to the explanation, then triple tap on the phone to continue.
5. Open or restart the phone app, scan the QR code with the phone, and wait for the HoloLens to say Phone QR detected.
6. Follow the posture instruction and example image, then triple tap to start. When the HoloLens says Triple tap detected. Hold still., hold still briefly.
7. Before each task, the task name appears. Listen to the explanation, then triple tap on the phone to continue.
8. A short practice may appear before each task. Practice is not logged.
9. Before the first main trial of each task, the HoloLens announces Main Placement Task Starts Now, Main Rotation Task Starts Now, or Main Scaling Task Starts Now.
10. Match the target, then triple tap to submit.
11. After one condition ends, complete the break/questionnaire step, then triple tap to continue to the next condition.

3. Condition Summary

Macro - Near Head

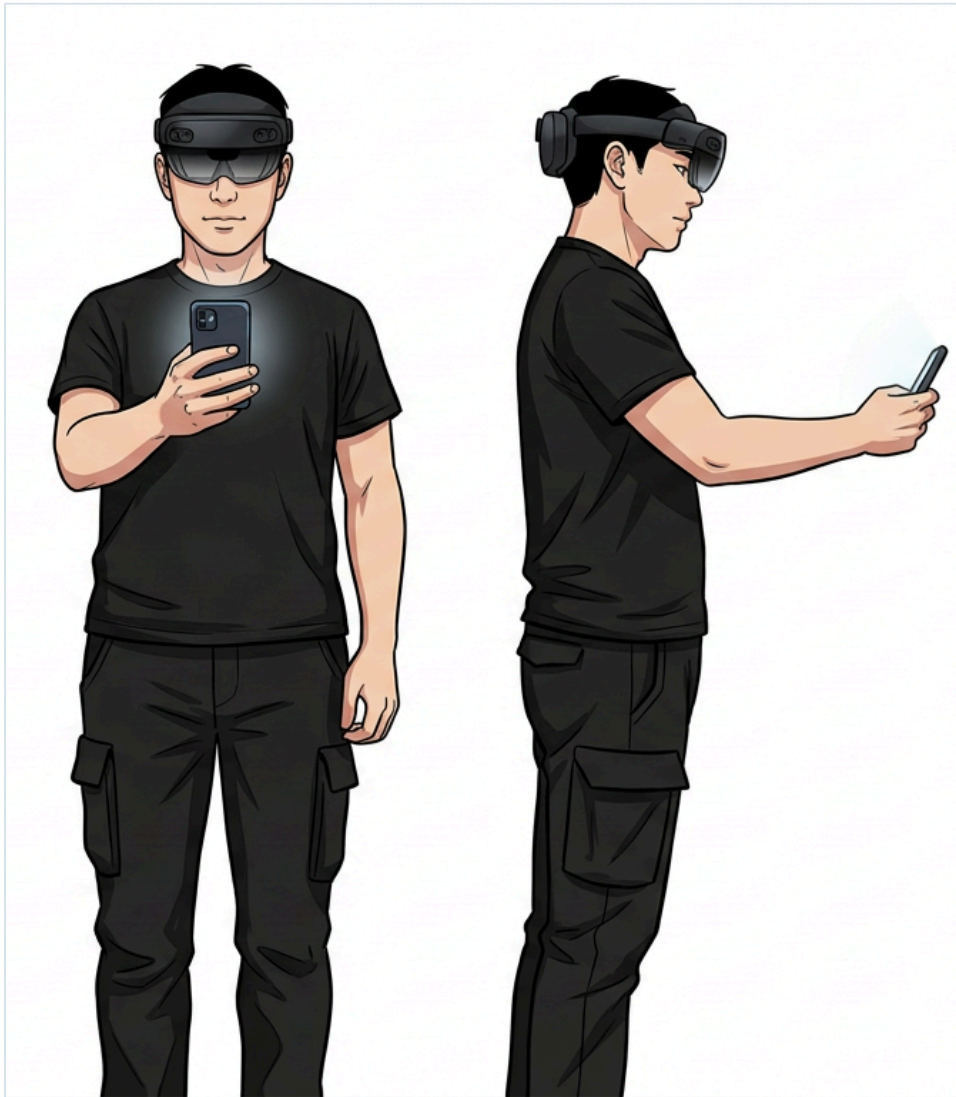
- Use larger arm and hand movements.
- Keep your hand near your head.

Spoken script (EN)

In this condition, keep your hand near your head.

You will use larger phone and arm movements to control the proxy hand.

After this explanation, triple tap to continue.



Macro - Side Of Body

- Use larger arm and hand movements.
- Keep your hand at the side of your body.
- In this condition, the input is remapped.
- Phone left/right movement stays mapped to proxy-hand left/right movement.
- Phone up/down movement controls proxy-hand forward/back movement.
- Phone forward/back movement controls proxy-hand up/down movement.

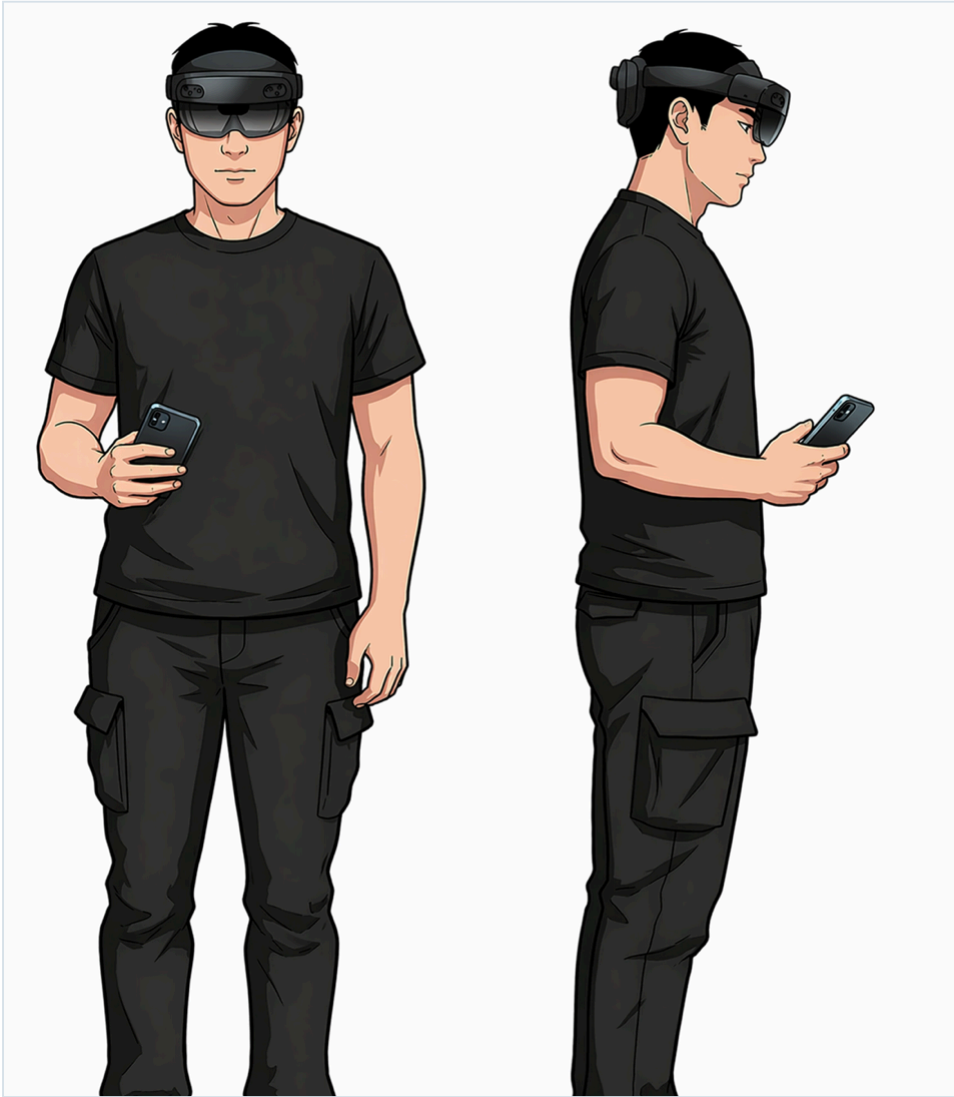
Spoken script (EN)

In this condition, keep your hand at the side of your body.

You will use larger phone and arm movements.

The movement is changed here: left and right stay left and right, up and down move forward and back, and forward and back move up and down.

After this explanation, triple tap to continue.



Micro - Near Head

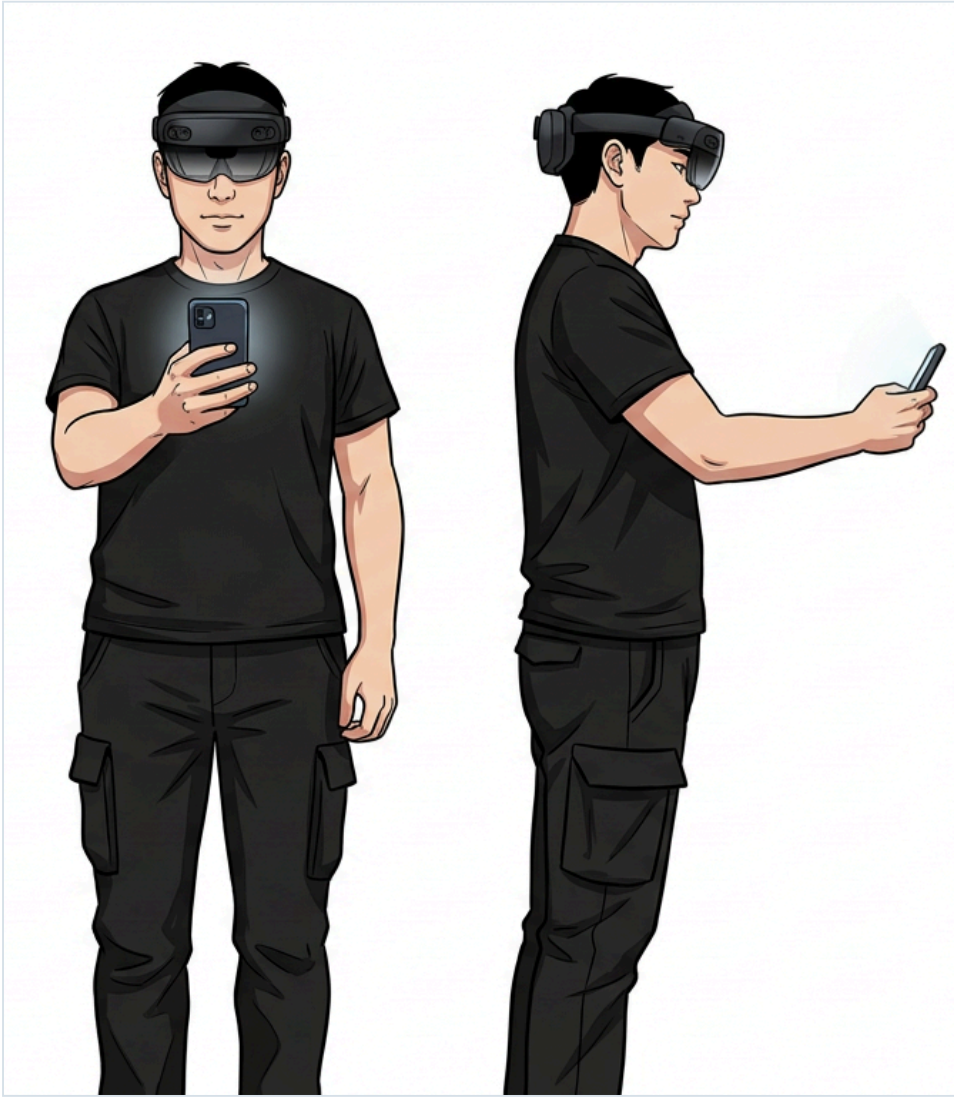
- Keep your hand near your head.
- Use phone screen swipes for fine control.

Spoken script (EN)

In this condition, keep your hand near your head.

Try to keep your arm still, and use small phone swipes for fine control.

After this explanation, triple tap to continue.



Micro - Side Of Body

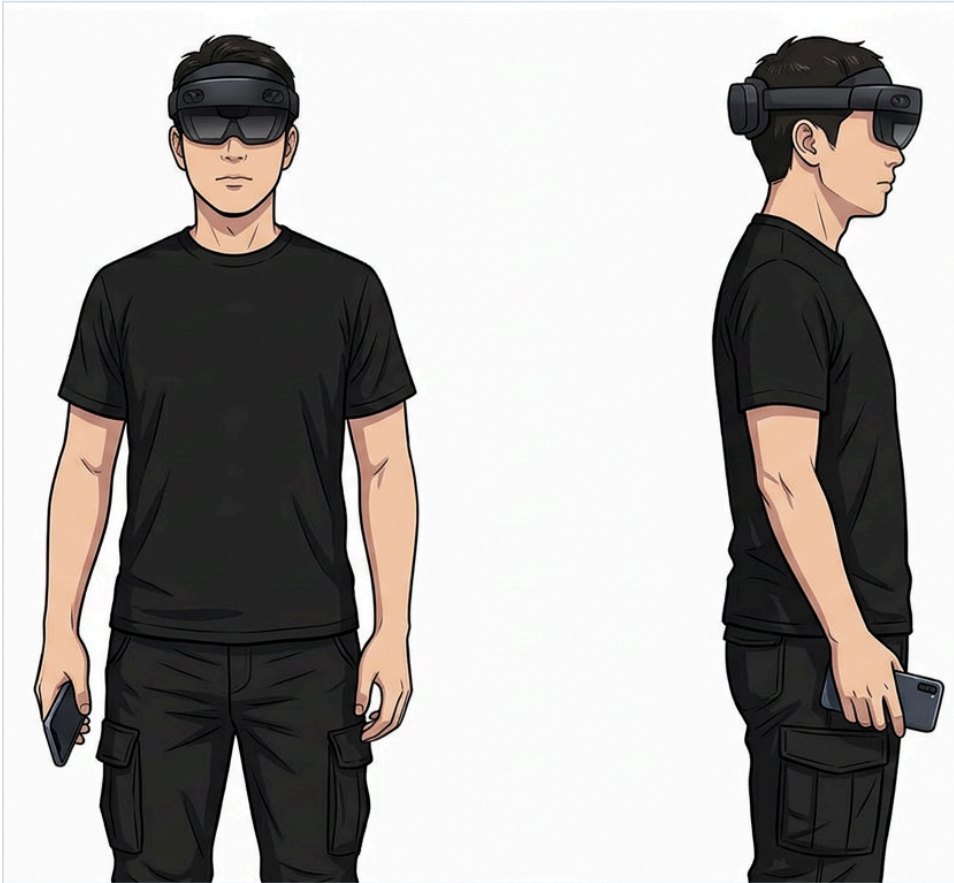
- Keep your hand at the side of your body.
- Use phone screen swipes for fine control.

Spoken script (EN)

In this condition, keep your hand at the side of your body.

Try to keep your arm still, and use small phone swipes for fine control.

After this explanation, triple tap to continue.



4. Task Summary

Placement

- Move the tool to the target position.
- In Micro, swipe on the phone to move the tool left/right and up/down.
- If needed, double tap to switch the movement plane for forward/back adjustment.
- After matching the target position, submit with triple tap.

Spoken script (EN)

In this task, move the tool to the target position.
Try to place it as close to the target as possible.
When you are done, triple tap to submit.

Micro add-on:

In micro placement, tap once to grab and tap again to release.
Swipe to move the tool. Double tap if you need to change the movement plane.

Rotation

- Match the target orientation.
- In Micro, left/right swipes control yaw, and up/down swipes control roll.
- If needed, double tap to switch the vertical swipe axis between roll and pitch.
- After matching the target orientation, submit with triple tap.

Spoken script (EN)

In this task, match the tool's angle to the target.
You do not need to match the position exactly. Focus on the rotation.
When you are done, triple tap to submit.

Micro add-on:

In micro rotation, swipe left and right to turn the tool.
Swipe up and down for another rotation direction. Double tap to switch that direction.

Scaling

- Match the target size.
- In Micro, up/down swipes make the tool larger or smaller.
- After matching the target size, submit with triple tap.

Spoken script (EN)

In this task, match the tool's size to the target size.
Make the tool as close to the target size as possible.
When you are done, triple tap to submit.

Micro add-on:

In micro scaling, swipe up to make the tool smaller, and swipe down to make it bigger.

5. Notes

- In Macro, you control the task mainly with arm and hand movement.
- In Micro, you keep the instructed hand posture and use phone swipes for fine adjustments.
- If the proxy hand is close enough to an object, it can grab the object.
- In Macro, the object is held while you keep touching the screen, and it is released when you let go.
- In Micro, a single tap grabs the object, and another tap releases it.
- If the phone app is restarted for a new condition, scan the QR code with the phone again before matching the posture.
- Try not to change posture too much while the posture instruction image is shown and while the HoloLens asks you to hold still.
- If QR scanning fails, Phone QR detected. is not announced, or the display/input seems wrong, tell the facilitator immediately.

6. Most Important

- Before starting, enter your Participant ID and scan the HoloLens workspace QR code.
- For every condition, listen to the condition explanation, triple tap to continue, scan the QR code with the phone, and wait for Phone QR detected. before posture matching and triple tap.
- Keep your hand in the correct condition posture: Near Head or Side Of Body.
- When the HoloLens says Triple tap detected. Hold still., keep still briefly.
- In Macro, you hold an object while touching the screen. In Micro, you tap to grab and tap again to release.
- In Micro, use phone swipes to control the task.

- In Macro - Side Of Body, phone left/right stays left/right, phone up/down drives forward/back, and phone forward/back drives up/down.
- Submission is always triple tap for every task.